20 Lecture 20 (March 3, 2025, est.): Frobenius type and lifting

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We continue over an algebraically closed field k of characteristic $p > 0$. Our categories are semisimple unless otherwise specified. The Frobenius functor takes values in $\mathcal{C}^{(1)} \boxtimes \mathsf{Ver}_p$, with the scalar twist of Lecture 19.

20.1 Frobenius on the Verlinde category

Assume $p > 2$, and put $\varepsilon = L_{p-1}$. Recall the fusion rules

$$L_r \otimes L_s = \bigoplus_{i=1}^{\min(r,s,p-r,p-s)} L_{|r-s|+2i-1}, \qquad \varepsilon \otimes L_s = L_{p-s}.$$

In particular $\varepsilon^{\otimes 2} = \mathbf{1}$, and $\langle \varepsilon \rangle \simeq \mathrm{SVec}$. The subcategory Ver_p^+ consists of sums of the L_s with s odd.

20.1 Proposition. *Using the prime-field identification* $\mathsf{Ver}_p^{(1)} \simeq \mathsf{Ver}_p$*, one has*

$$\mathrm{Fr}(L_s) = \begin{cases} \mathbf{1} \boxtimes L_s, & s \text{ odd}, \\ \varepsilon \boxtimes L_{p-s}, & s \text{ even}. \end{cases}$$

Proof. Set $x = [L_2]$ in $K_0(\mathsf{Ver}_p)$. The recurrence $[L_{s+1}] = x[L_s] - [L_{s-1}]$, together with $[L_0] = [L_p] = 0$, gives

$$x^p = -2[\varepsilon] \quad \text{in } K_0(\mathsf{Ver}_p) \otimes k.$$

One way to verify this polynomial identity is to put $[L_s] = U_{s-1}(x/2)$, where $U_0 = 1$, $U_1 = 2t$, and $U_{m+1} = 2tU_m - U_{m-1}$, and reduce modulo p and $U_{p-1}(x/2)$.

Now $\mathrm{FPdim}(L_2) = 2\cos(\pi/p) < 2$. Preservation of this dimension implies that $\mathrm{Fr}(L_2)$ is simple: every nonzero simple summand has Frobenius–Perron dimension at least one. Write it as $L_a \boxtimes L_b$. The identity of Lemma 19.5 says $b[L_a] = -2[L_{p-1}]$ in $K_0(\mathsf{Ver}_p) \otimes k$. Since $1 \le b < p$, comparison of basis coefficients gives $a = p - 1$ and $b = p - 2$. Finally apply Fr to $L_2 \otimes L_s = L_{s-1} \oplus L_{s+1}$ and use induction on s. The displayed fusion rules verify the two parity cases and give the formula. ∎

20.2 Frobenius type

20.2 Definition. The *Frobenius type* of $\mathcal{C}$ is the smallest fusion subcategory $\mathcal{A} \subseteq \mathsf{Ver}_p$ such that

$$\mathrm{Fr}(\mathcal{C}) \subseteq \mathcal{C}^{(1)} \boxtimes \mathcal{A}.$$

Equivalently, it is generated by all L_r occurring as a second factor in some simple summand of some $\mathrm{Fr}(X)$.

For $p > 3$, the possibilities are Vect, SVec, Ver_p^+, and Ver_p. To see why this list is exhaustive, a subcategory containing any L_r other than $\mathbf{1}$ or ε contains L_3 as a summand of $L_r \otimes L_r^*$. The fusion rules show that L_3 generates all the odd-indexed simples. Adding any even-indexed simple then generates ε and hence all of Ver_p.

20.3 Example. Both Vect and SVec have Frobenius type Vect; for the odd line π and odd p, the p-cycle acts trivially on $\pi^{\otimes p}$. Proposition 20.1 shows that Ver_p has type Ver_p^+. In particular the rank-two category Ver_5^+ has type Ver_5^+, as computed in the previous lecture. When $p = 2$, $\mathsf{Ver}_2 = $ Vect; when $p = 3$, $\mathsf{Ver}_3 = $ SVec and $\mathsf{Ver}_3^+ = $ Vect.

20.4 Proposition. *For an exact symmetric tensor functor $F : \mathcal{C} \to \mathcal{D}$, there is a natural monoidal isomorphism*

$$\mathrm{Fr}_{\mathcal{D}} \circ F \simeq (F^{(1)} \boxtimes \mathrm{id}) \circ \mathrm{Fr}_{\mathcal{C}}.$$

Consequently the Frobenius type of $\mathcal{C}$ is contained in that of $\mathcal{D}$.

Proof. The symmetric structure of F identifies the cyclic actions on $F(X^{\otimes p})$ and $F(X)^{\otimes p}$. In the semisimple construction, this identification is compatible with taking the multiplicity spaces and with the quotient of $\mathrm{Rep}_k(C_p)$ by negligible maps. It gives the stated isomorphism. Moreover an exact tensor functor is faithful, so it sends no nonzero object to zero. Thus every second-factor summand occurring before applying F still occurs afterwards; it must belong to the type of $\mathcal{D}$. ∎

In particular, a category admitting a super fiber functor has type Vect. If $\mathcal{C}$ itself has type Vect, its Frobenius functor lands in $\mathcal{C}^{(1)}$ and

$$[\mathrm{Fr}(X)] = [X]^p \quad \text{in } K_0(\mathcal{C}) \otimes_{\mathbb{Z}} k. \tag{20.1}$$

20.3 Scalar extension and split categories

For the discussion of lifting, temporarily let F be an arbitrary field. A finite semisimple F-linear category is *split* if $\mathrm{End}(S) = F$ for every simple object S. Equivalently, every simple remains simple after extension to an algebraic closure. Scalar extension is formed by extending morphism spaces and taking the additive idempotent completion; we write $\mathcal{C} \otimes_F F'$ for a field extension F'/F.

For example, the real two-dimensional representation of C_3 in which a generator acts by rotation through $2\pi/3$ is simple over $\mathbb{R}$. Its endomorphism algebra is $\mathbb{C}$, and after extension to $\mathbb{C}$ it splits into two one-dimensional characters. Thus $\mathrm{Rep}_{\mathbb{R}}(C_3)$ is not split.

20.4 The lifting theorem

For a symmetric fusion category, define its *global dimension* by

$$\dim(\mathcal{C}) = \sum_{S \text{ simple}} \dim(S)^2 \in k.$$

We call $\mathcal{C}$ *nondegenerate* when this scalar is nonzero. Here this word concerns global dimension; it does not mean that the braiding has trivial symmetric center.

Return to an algebraically closed field k of characteristic p, which is in particular perfect, and let $W(k)$ be its ring of Witt vectors. This is a complete discrete valuation ring of characteristic zero with residue field k. A split lifting of a split fusion category $\mathcal{C}$ to $W(k)$ consists of a rigid monoidal $W(k)$-linear category with finite free morphism modules, the same simple labels and fusion multiplicities, and reduction modulo p equivalent to $\mathcal{C}$. Extending to the fraction field gives a characteristic-zero fusion category.

20.5 Theorem (Etingof–Nikshych–Ostrik)**.** *A nondegenerate fusion category over an algebraically closed field k lifts to $W(k)$, uniquely up to equivalence with prescribed reduction. A braiding, and in particular a symmetry, lifts as well. Lifting is faithful on equivalence classes of nondegenerate categories.*

These are the lifting and faithfulness results of [ENO05, Section 9]. They are input theorems here; their proof uses vanishing of deformation cohomology. The completeness and residue-field conditions on the base ring are essential to this formulation: the theorem is not an assertion about every local ring with residue field k.

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